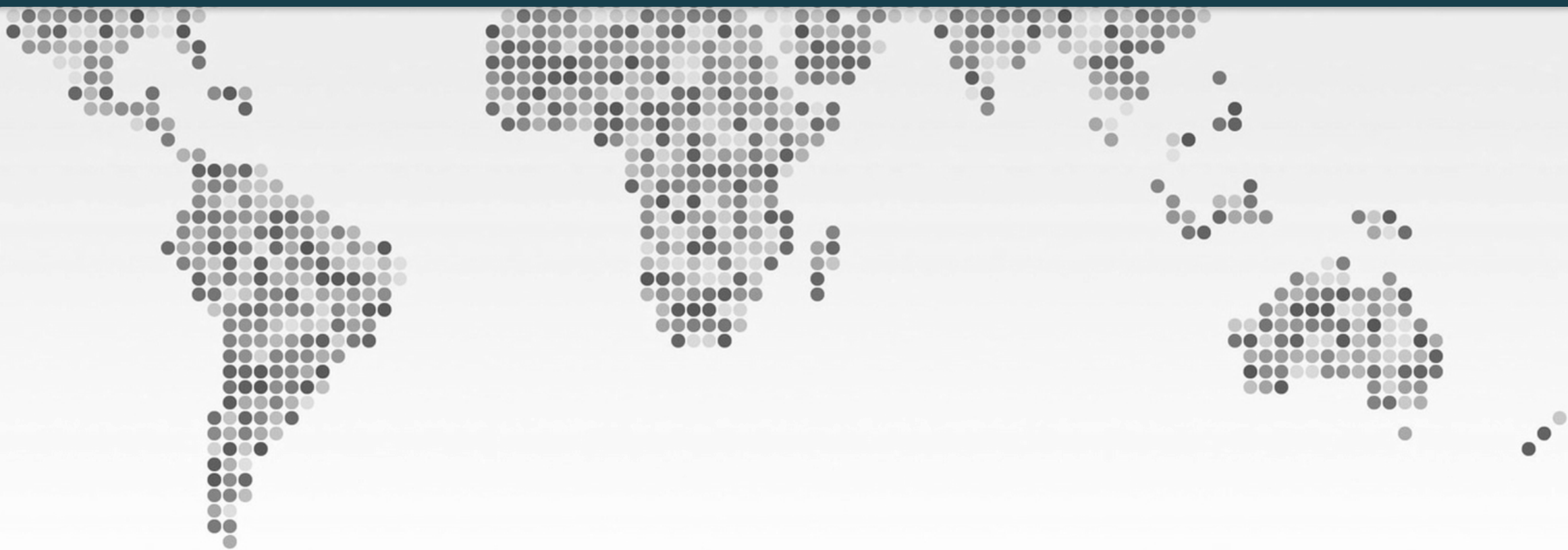


Software Engineering for Artificial Intelligence

Data and Feature Engineering



Software Engineering for Artificial Intelligence: Data and Feature Engineering

Data and Feature Engineering: Why?

There cannot be artificial intelligence without data. There cannot be accurate artificial intelligence without *good* data.

Data Governance. The management process that curates the availability, usability, integrity, and safety of the data.

Historically speaking, data governance has received increasing attention because of multiple reasons connected to financial, environmental, and governmental regulations.

Nowadays, data represent one of the major values of companies, organizations, and governments. Implementing strategies to collect, curate, and store data is therefore a urgent need that pertain to the definition of internal and global standards.

The main focus of data governance is concerned with data quality —> good data have higher value and can be used to advance the state of the practice by means of AI!

Data Quality. The set of processes that describe the accuracy, completeness, and consistency of the data collected.

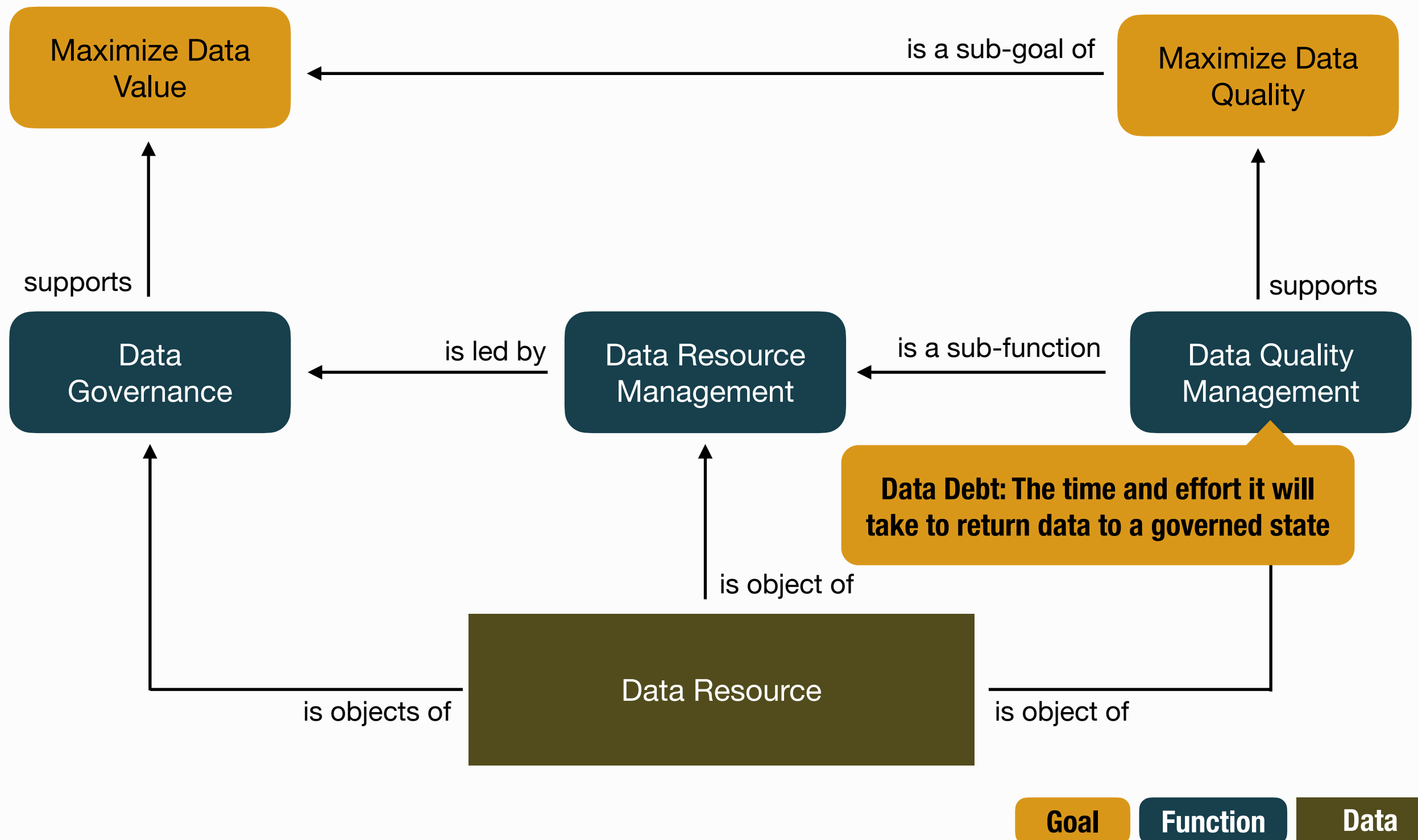
To enable data quality, we refer to data engineering.

Data Engineering. The set of techniques and algorithms that enable the extraction, analysis, and preparation of data. It is the backbone of artificial intelligence.

Software Engineering for Artificial Intelligence: Data and Feature Engineering

Data and Feature Engineering: Why?

Let's take a closer look at why data governance is strongly connected to data quality.



Software Engineering for Artificial Intelligence: Data and Feature Engineering

Data and Feature Engineering: Why?

Data debt, uhm? A term which is borrowed from the agile software development vocabulary and revolves around the application of sub-optimal data quality and management processes that lead data not to be meaningful enough.

Data debt is the major obstacle to data-driven organizations. This includes various sources of problems with data.

Structural debt

Quality issue with the design of a table, column, or view. Examples are the improperly named column, insufficiently normalized data, etc.

Data quality debt

Quality issue with the consistency or usage of data values. Examples are the duplicated business key value, corrupted data, etc.

Integrity debt

Quality issue with whether a referenced row exists within another table and that a row which is no longer needed is (soft) deleted appropriately, e.g., null foreign keys.

Architectural debt

Quality issue with how external programs interact with a data source. Examples are inappropriate security access control, inappropriate encapsulation, etc.

Documentation debt

Quality issue with with any supporting documents, including models. Examples are inconsistent or outdated information.

Functional debt

Quality issue with execution aspects within a data source, such as stored procedures, stored functions, and triggers.

While some of these issues can be tackled with good database skills, some are concerned with the preprocessing steps during the data extraction phase.

Software Engineering for Artificial Intelligence: Data and Feature Engineering

Data and Feature Engineering: Key figures

Most of the work required to deal with data debt is done by the so-called *data engineer*, who is responsible of all processes concerned with data extraction and preprocessing. More particularly, we recognize three types of data engineers:

Architect

Typically, this term indicates that a data engineer is the responsible of all the steps of the data engineering pipeline.

Database engineer

As the name suggests, the role involves all the modeling, development, and maintenance activities of the databases of an organization.

Pipeline engineer

This is a complementary role, which is involved in the steps of data cleaning and collection, namely the steps before storing information in a database.

As you see, there is no *data scientist* involved in the data engineering process.

Indeed, the data scientist role does not revolve around the collection of data, but their analysis. The aim of a data scientist is to make sense of the collected data.

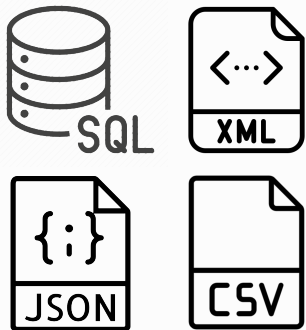
There is, however, a strong connection between data engineers and scientists. They have to work closely together, as certain information might require transformations before being stored. That is why we refer to both data and feature engineering!

Data vs Feature. A data is any element available to solve a problem, a feature is a characteristic of the problem that may be composed from data.

Software Engineering for Artificial Intelligence: Data and Feature Engineering

Data and Feature Engineering: Which Data Can You Mine?

Potentially, everything is mineable! Of course, different types of data would require different mining instruments.



Structured data

Structured data are tabular sources which have a well-defined structure. These are the easiest to mine, as they are already organized and often available.

See here: https://en.wikipedia.org/wiki/List_of_datasets_for_machine-learning_research



Unstructured data

Unstructured data are represented by any other form of file which does not follow a defined structure. These are the hardest to mine, as they require ad-hoc parsers.

See here: <https://index.quantumstat.com>

Quite often, data are available in hybrid forms. In those cases, the hardness of the data collection might be even higher (or lower, it depends).

Independently from the types of data you can mine, it is worth classifying the various *sources* of information that may be used to devise artificial intelligence systems.

Based on the problem of interest, you might want to consider different perspectives.

Software Engineering for Artificial Intelligence: Data and Feature Engineering

Data and Feature Engineering: Which Data Can You Mine?

To better explain the meaning of *data sources*, let's consider the problem of understanding the evolution of source code in a software system.

Let's suppose to have this piece of code:

```
void method(...){  
    noFirewall = new JRadioButton("no firewall");  
    socksFirewall = new JRadioButton("no SOCKS 4/5 Firewall");  
    webProxy = new JRadioButton("HTTP web proxy");  
}
```

Now, answer these questions:

Q1. How many objects are created in the method?

Structural information

Q2. What does this method do?

Textual information

Q3. When was the method introduced?

Historical information

Q4. What is the execution time of the method?

Dynamic information

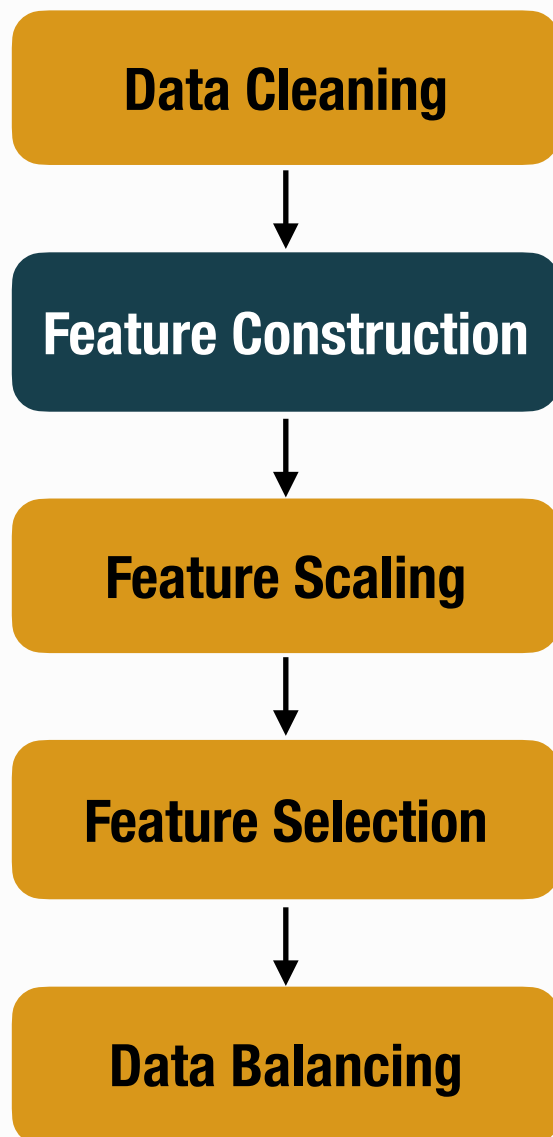
Depending on the context, you might have additional sources that you can consider. For instance, multimedia data, telemetry data, and so on.

Software Engineering for Artificial Intelligence: Data and Feature Engineering

Data and Feature Engineering: Toward Data Quality and Feature Engineering

From a practical perspective, there is not very much to say on the tools you can use to mine data: there exist a variety of instruments, but their use depends on what you would like to mine.

Much more can instead be said with respect to the follow up steps to conduct in order to make data qualitative enough to build reliable artificial intelligence methods.



First, data can be noisy and/or incomplete. The data cleaning procedures aim at normalizing the data extracted.

Once you have data, you can start making sense of them. The feature construction is not a strict data quality process, but without it you could not have any of the subsequent steps —> without features, you could not scale them.

The subsequent steps of feature scaling and selection aim at working on the features by putting them within specified scales and removing redundant variables that might bias the artificial intelligence models.

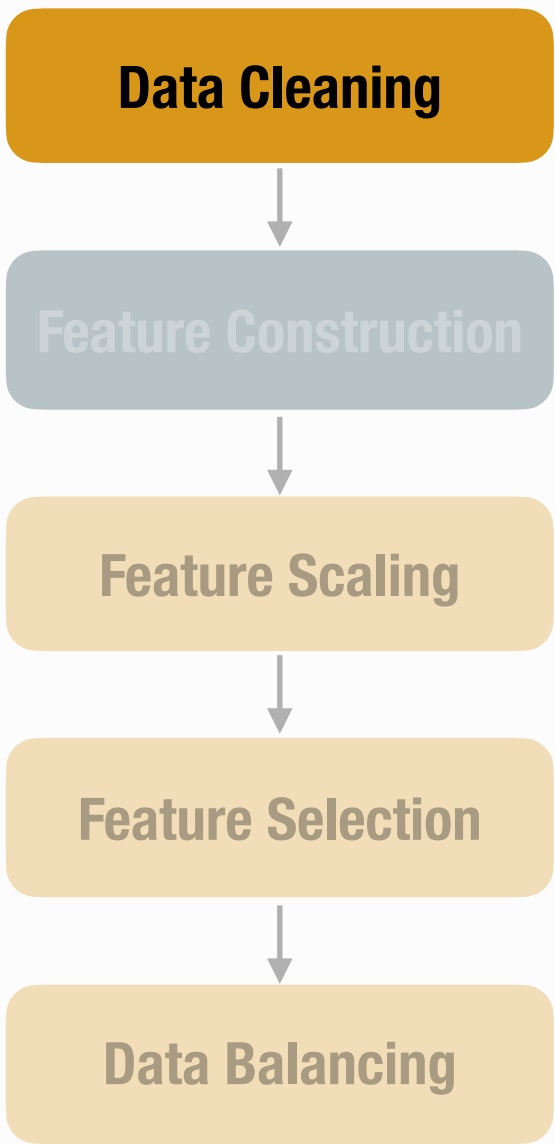
Finally, data balancing considerations are crucial to let artificial intelligence methods to properly learn the features and characteristics of the problem of interest.

Software Engineering for Artificial Intelligence: Data and Feature Engineering

Data and Feature Engineering: Toward Data Quality and Feature Engineering

From a practical perspective, there is not very much to say on the tools you can use to mine data: there exist a variety of instruments, but their use depends on what you would like to mine.

Much more can instead be said with respect to the follow up steps to conduct in order to make data qualitative enough to build reliable artificial intelligence methods.



To make these steps more practical, let's consider the problem of predicting whether an email should be marked as spam or not.

is_spam	domain	subject				
NO	<u>unisa.it</u>	[SE4AI] Exam				
NO	<u>gmail.com</u>	[SE4AI] Exam	-	180	black	yes
YES	<u>live.it</u>	Business Interest!	-	222	black	no
YES	<u>spam.it</u>	How to get money!	3	492	-	
NO	<u>unisa.it</u>	Appointment request	0	300	black	no

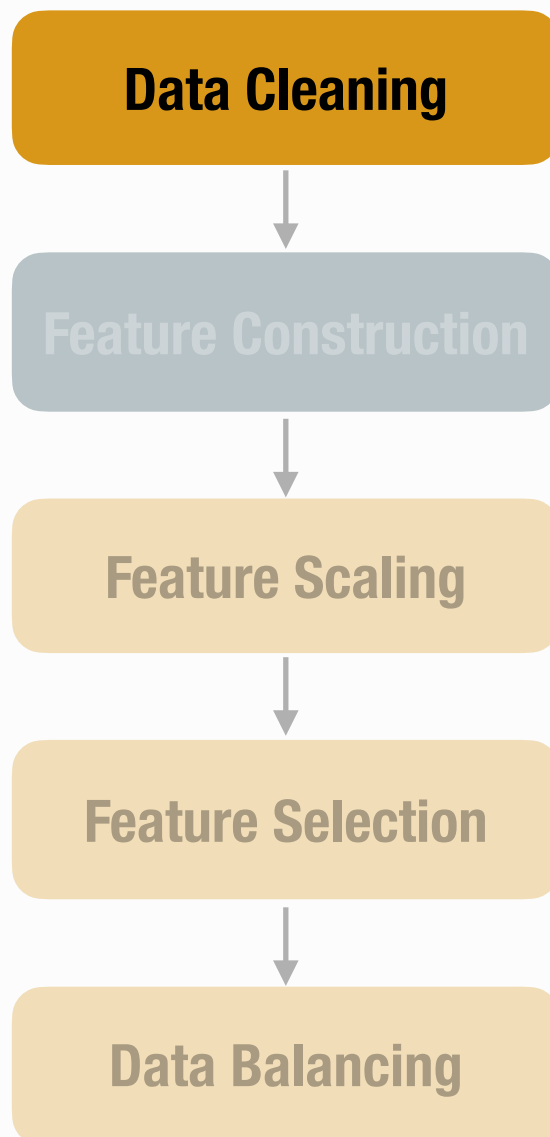
You extracted a bunch of data potentially useful for the problem, but you realize that some of those data could not be extracted (e.g., because of unavailability of information). What would you do?

Software Engineering for Artificial Intelligence: Data and Feature Engineering

Data and Feature Engineering: Toward Data Quality and Feature Engineering

From a practical perspective, there is not very much to say on the tools you can use to mine data: there exist a variety of instruments, but their use depends on what you would like to mine.

Much more can instead be said with respect to the follow up steps to conduct in order to make data qualitative enough to build reliable artificial intelligence methods.



To make these steps more practical, let's consider the problem of predicting whether an email should be marked as spam or not.

Solution 1. Remove the rows affected by missing data.

Problem 1. What is we do not have many observations?

Solution 2. Remove the columns affected by missing data.

Problem 2. What is the column represents a relevant feature?

So, what?

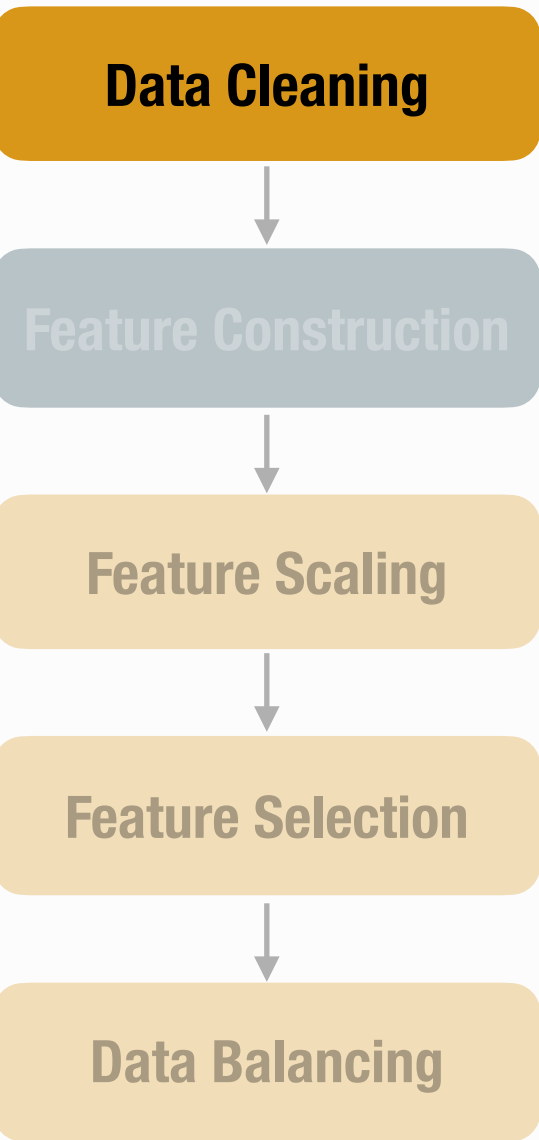
Data Imputation. Methods to deal with incomplete data, aiming at estimating them based on the data available or based on logical implications.

Software Engineering for Artificial Intelligence: Data and Feature Engineering

Data and Feature Engineering: Toward Data Quality and Feature Engineering

From a practical perspective, there is not very much to say on the tools you can use to mine data: there exist a variety of instruments, but their use depends on what you would like to mine.

Much more can instead be said with respect to the follow up steps to conduct in order to make data qualitative enough to build reliable artificial intelligence methods.



Data Imputation. Methods to deal with incomplete data, aiming at estimating them based on the data available or based on logical implications.

Statistical imputation. The use of statistical operators to estimate the value of a missing attribute.

	col1	col2	col3	col4	col5		col1	col2	col3	col4	col5	
0	2	5.0	3.0	6	NaN	mean()	0	2.0	5.0	3.0	6.0	7.0
1	9	NaN	9.0	0	7.0		1	9.0	11.0	9.0	0.0	7.0
2	19	17.0	NaN	9	NaN		2	19.0	17.0	6.0	9.0	7.0

For instance, the mean/median operator. Be careful though: this is possible for numeric data only!

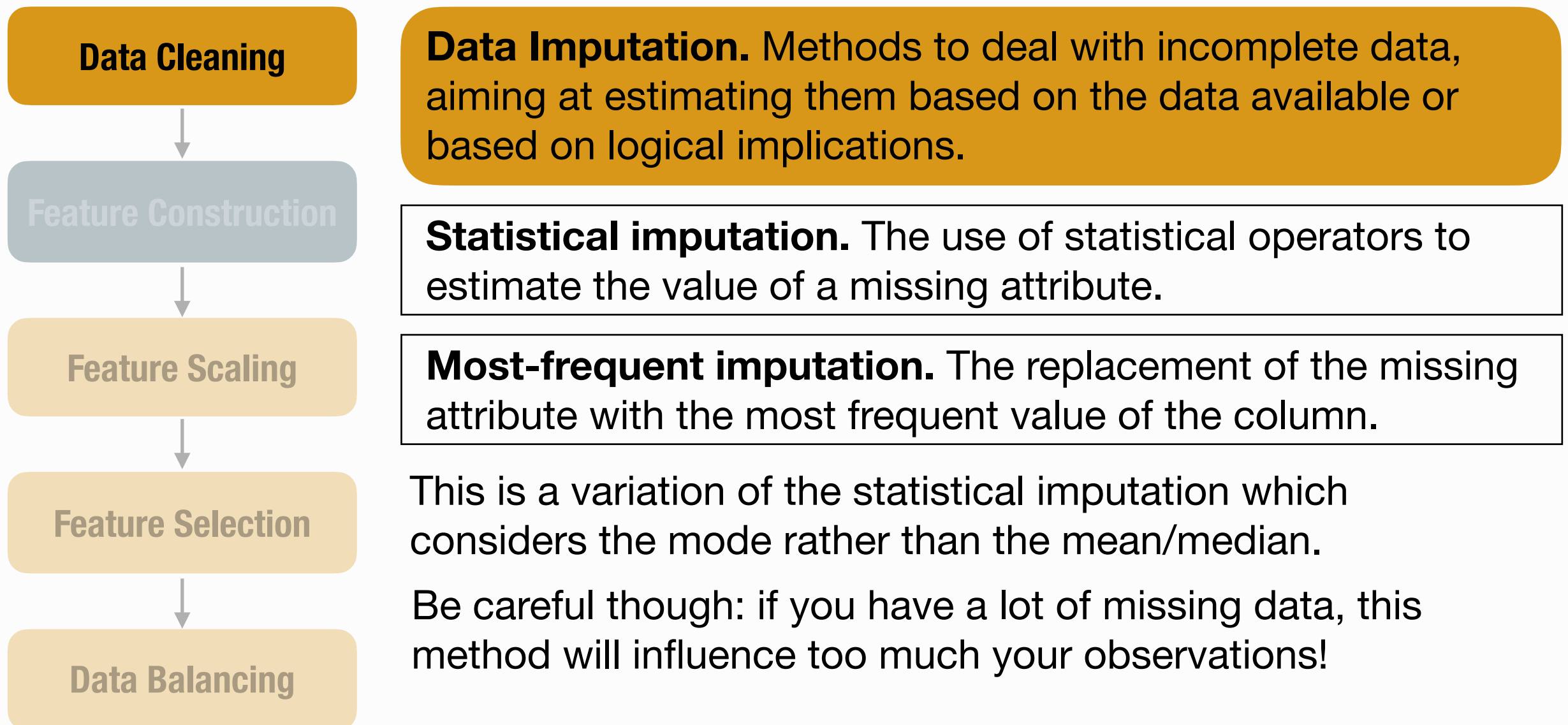
In addition, the method does not care of data uncertainty.

Software Engineering for Artificial Intelligence: Data and Feature Engineering

Data and Feature Engineering: Toward Data Quality and Feature Engineering

From a practical perspective, there is not very much to say on the tools you can use to mine data: there exist a variety of instruments, but their use depends on what you would like to mine.

Much more can instead be said with respect to the follow up steps to conduct in order to make data qualitative enough to build reliable artificial intelligence methods.

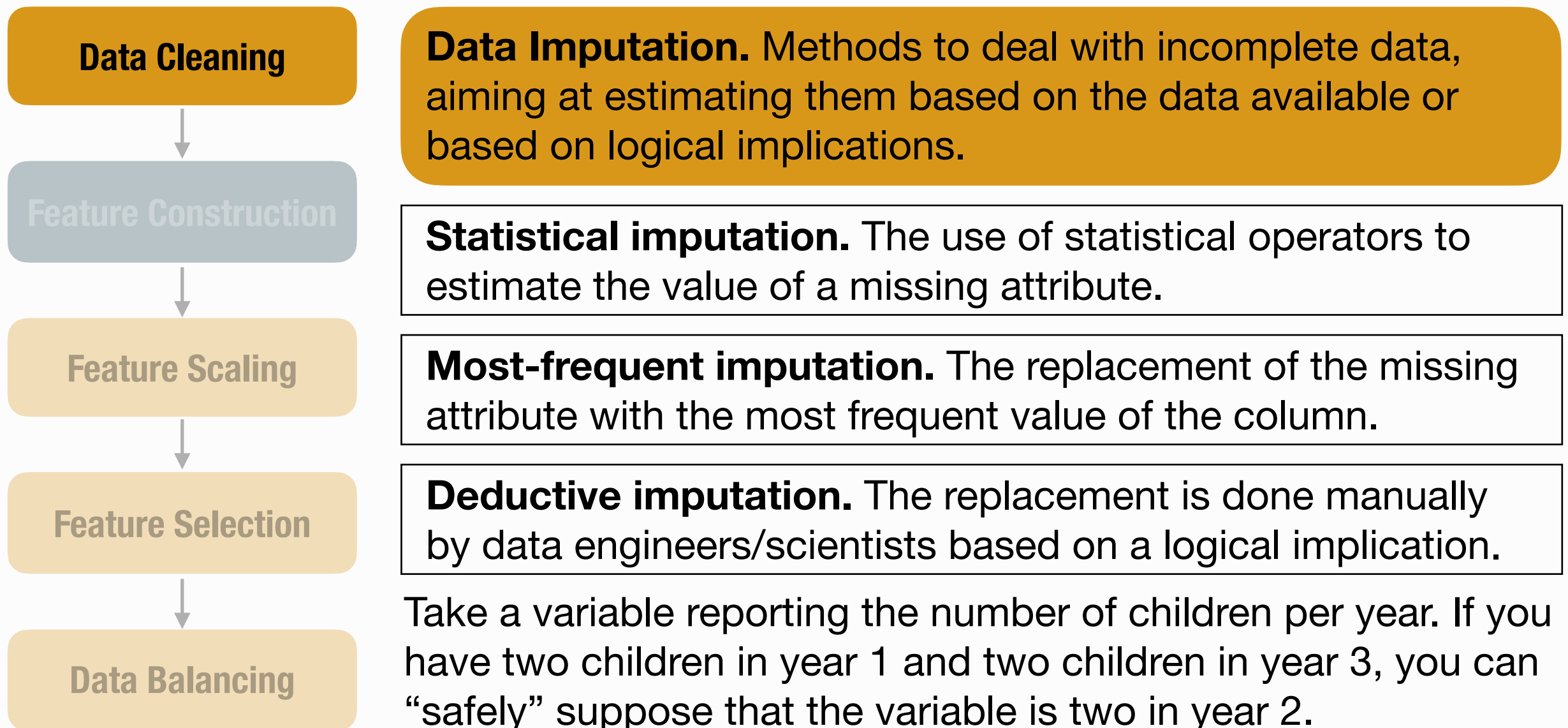


Software Engineering for Artificial Intelligence: Data and Feature Engineering

Data and Feature Engineering: Toward Data Quality and Feature Engineering

From a practical perspective, there is not very much to say on the tools you can use to mine data: there exist a variety of instruments, but their use depends on what you would like to mine.

Much more can instead be said with respect to the follow up steps to conduct in order to make data qualitative enough to build reliable artificial intelligence methods.

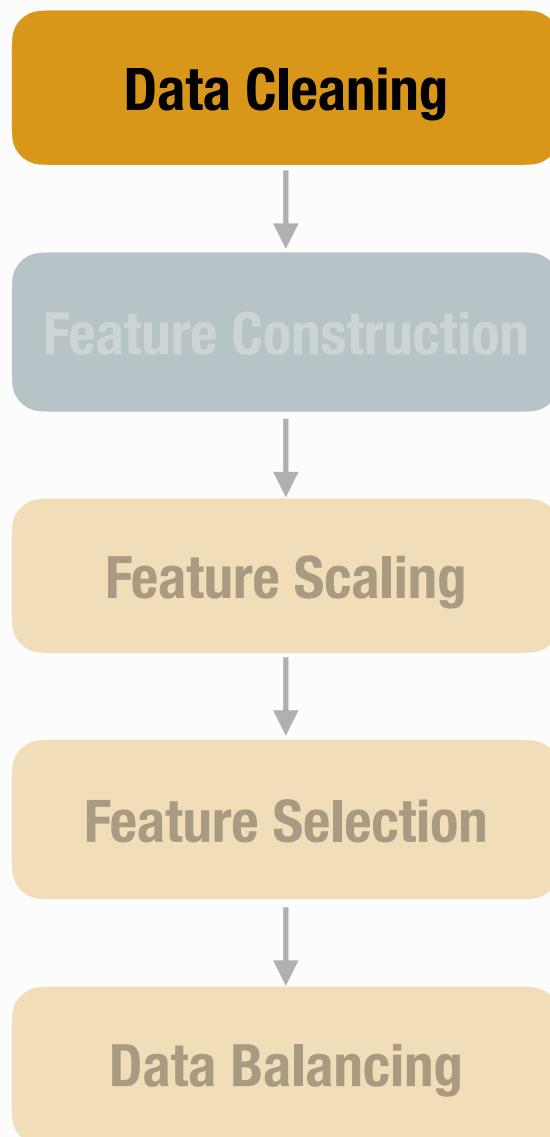


Software Engineering for Artificial Intelligence: Data and Feature Engineering

Data and Feature Engineering: Toward Data Quality and Feature Engineering

From a practical perspective, there is not very much to say on the tools you can use to mine data: there exist a variety of instruments, but their use depends on what you would like to mine.

Much more can instead be said with respect to the follow up steps to conduct in order to make data qualitative enough to build reliable artificial intelligence methods.



Data Imputation. Methods to deal with incomplete data, aiming at estimating them based on the data available or based on logical implications.

Q5. How would you select the method to use?

It depends on the extent of the problem. If this is limited, you might just drop columns/rows.

Otherwise, you might want to use more complex methods or even a combination of them.

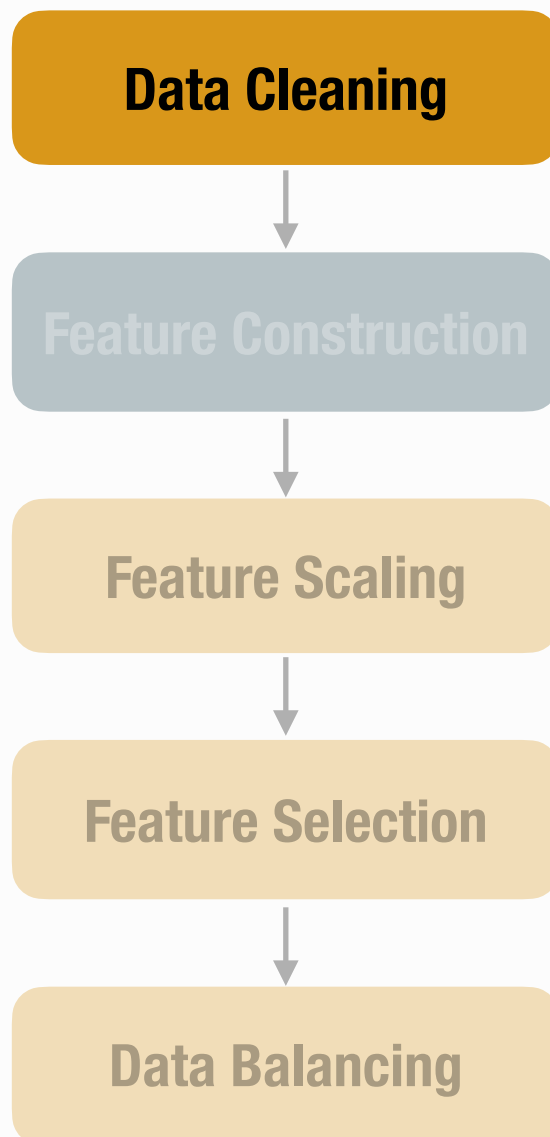
The most fearless folks might also consider the definition of supervised or semi-supervised learning mechanisms that can predict missing values based on the data available.

Software Engineering for Artificial Intelligence: Data and Feature Engineering

Data and Feature Engineering: Toward Data Quality and Feature Engineering

From a practical perspective, there is not very much to say on the tools you can use to mine data: there exist a variety of instruments, but their use depends on what you would like to mine.

Much more can instead be said with respect to the follow up steps to conduct in order to make data qualitative enough to build reliable artificial intelligence methods.



Data imputation is not the only matter to face when cleaning data:

- (1) Addressing formatting inconsistencies;
- (2) Removing redundant rows, if needed;
- (3) Removing redundant columns, if needed;
- (4) ...

Essentially, all the operations required to have a dataset which is the most reliable possible. All the steps depend on your knowledge of the dataset and your analysis skills.

Cleaning up your data is not the most glamorous of tasks, but it's an essential part of data wrangling. [...] Knowing how to properly clean and assemble your data will set you miles apart from others in your field.

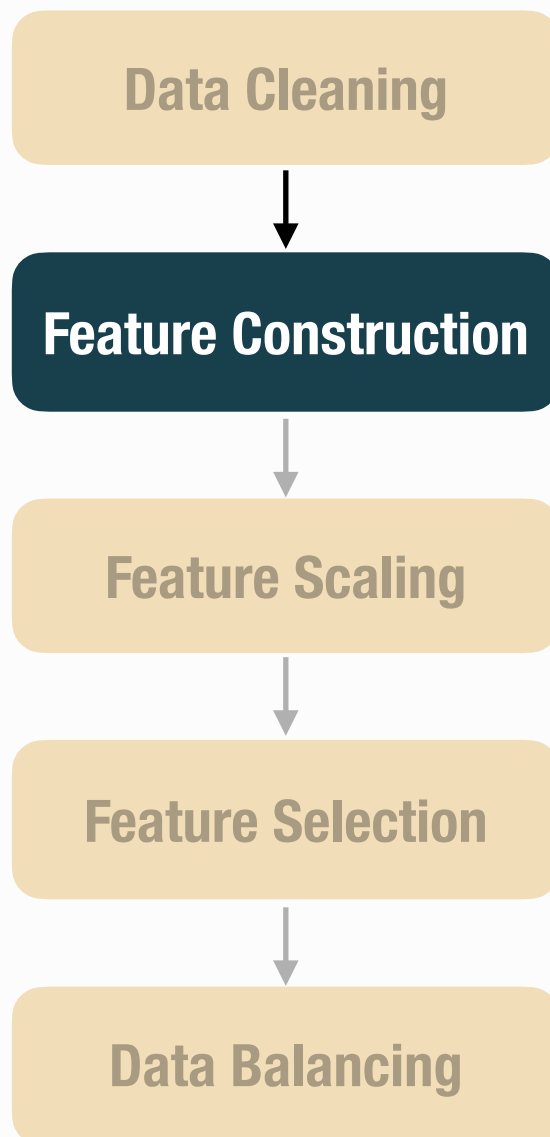
[J. Kazil & K. Jarmul - Data Wrangling in Python]

Software Engineering for Artificial Intelligence: Data and Feature Engineering

Data and Feature Engineering: Toward Data Quality and Feature Engineering

From a practical perspective, there is not very much to say on the tools you can use to mine data: there exist a variety of instruments, but their use depends on what you would like to mine.

Much more can instead be said with respect to the follow up steps to conduct in order to make data qualitative enough to build reliable artificial intelligence methods.



Feature Construction. The process of building intermediate features from the original data. The aim is to build more efficient features for a machine data mining task.

So far, you have “just” data. What do these data mean? How can they be used within an artificial intelligence algorithm?

This is the step where a data scientist takes the lead.

The goal is to understand how the data can be used to address the requirements of the artificial intelligence system.

While this is mainly a work of creativity, it might be guided by means of both manual and automatic analyses.

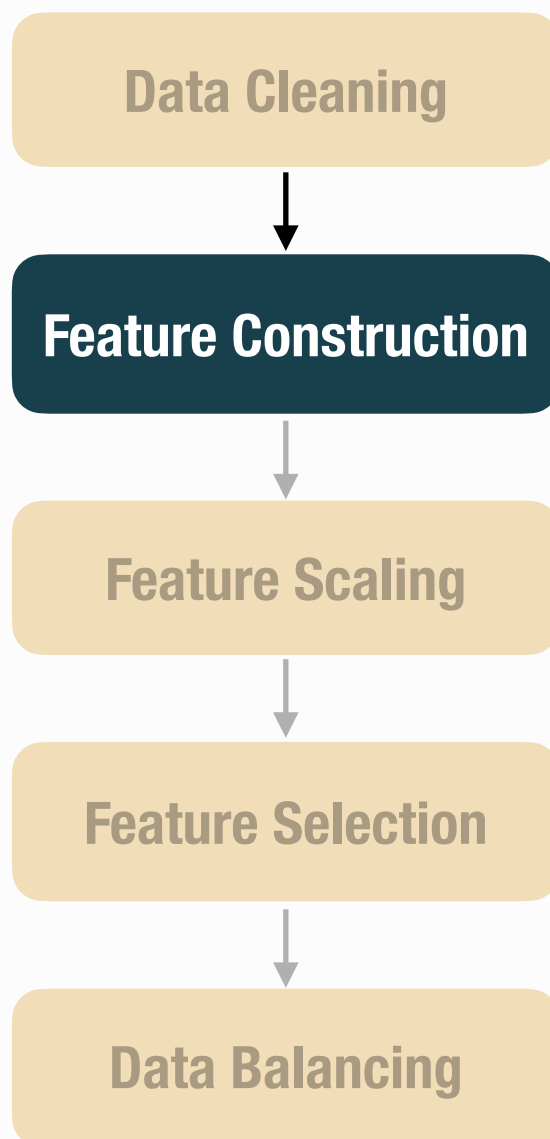
Manual analyses revolve around the understanding of the domain, automatic analyses around the understanding of data.

Software Engineering for Artificial Intelligence: Data and Feature Engineering

Data and Feature Engineering: Toward Data Quality and Feature Engineering

From a practical perspective, there is not very much to say on the tools you can use to mine data: there exist a variety of instruments, but their use depends on what you would like to mine.

Much more can instead be said with respect to the follow up steps to conduct in order to make data qualitative enough to build reliable artificial intelligence methods.



What does understanding of the *domain* mean?

You cannot automate an operation if you do not know how to do it manually. With artificial intelligence you typically try to let machines do something that is too costly for humans; however, humans know how to do that - otherwise, how can you hope that machines can do that?

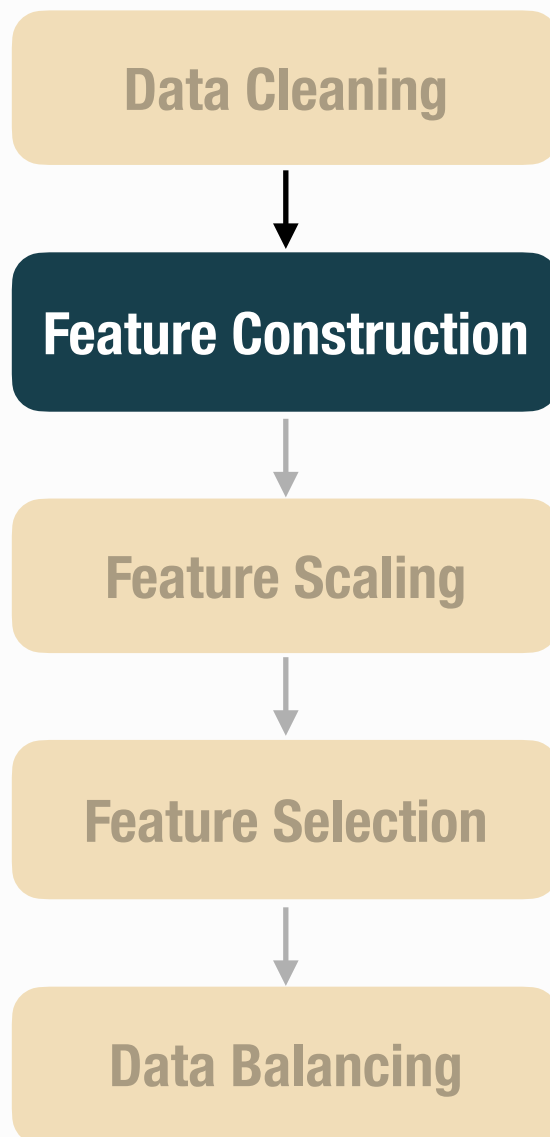
Hence, the data scientists first have the role of writing down notes on how the task can be done manually. In this way, they can understand the characteristics of the task and possibly trying to operationalize the operations that lead to performing it. As an example, just think at how Alexa can start your coffee machine: it just operationalizes what you are supposed to do it manually!

Software Engineering for Artificial Intelligence: Data and Feature Engineering

Data and Feature Engineering: Toward Data Quality and Feature Engineering

From a practical perspective, there is not very much to say on the tools you can use to mine data: there exist a variety of instruments, but their use depends on what you would like to mine.

Much more can instead be said with respect to the follow up steps to conduct in order to make data qualitative enough to build reliable artificial intelligence methods.



What does understanding of the *data* mean?

Starting from the data collected, a data scientist may explore them to measure how can be *reduced*. This would ease the job of feature construction, as s/he might work on a reduced set of possible features.

Dimensionality Reduction. Unsupervised transformation of data from a high-dimensional space into a low-dimensional space so that the low-dimensional representation retains some meaningful properties of the original data.

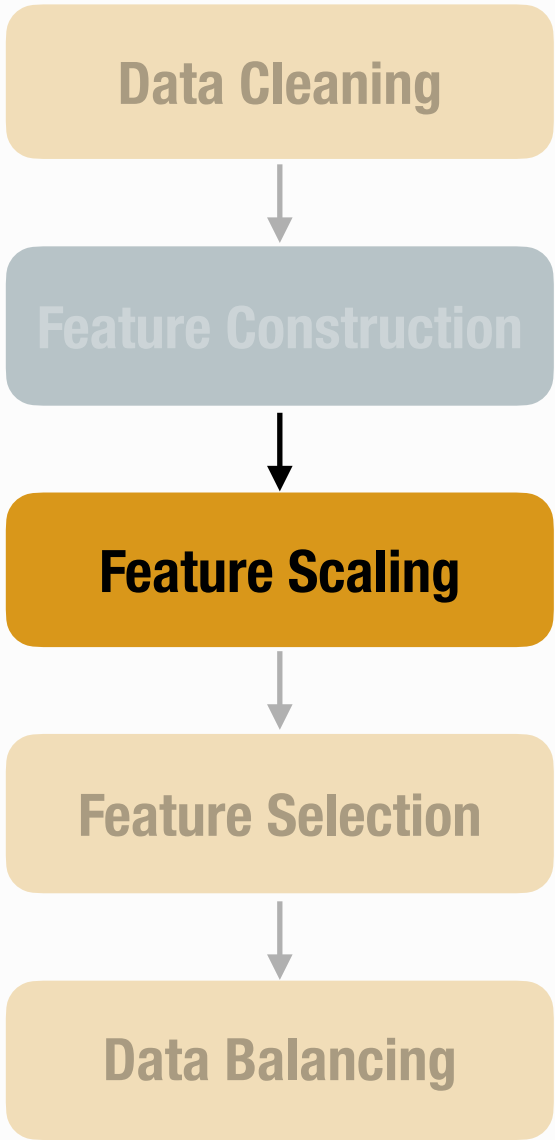
For the sake of time limitations, we cannot go deeper on dimensionality reduction. Yet, in the additional material you will find an article reporting on Principal Component Analysis, the most popular dimensionality reduction technique.

Software Engineering for Artificial Intelligence: Data and Feature Engineering

Data and Feature Engineering: Toward Data Quality and Feature Engineering

From a practical perspective, there is not very much to say on the tools you can use to mine data: there exist a variety of instruments, but their use depends on what you would like to mine.

Much more can instead be said with respect to the follow up steps to conduct in order to make data qualitative enough to build reliable artificial intelligence methods.



Let's reason again on the example shown before.

is_spam	domain	subject	attachments	length	text_color	contains_SE4AI
NO	<u>unisa.it</u>	[SE4AI] Ex				
NO	<u>gmail.com</u>	[SE4AI] Ex				
YES	<u>live.it</u>	Business Interest				
YES	<u>spam.it</u>	How to get money!	3	492	black	no
NO	<u>unisa.it</u>	Appointment request	0	300	black	no

The distribution of these two features is naturally different - the number of attachments is typically lower than the number of characters used in the message.

The different distribution of the two features might impact the performance of artificial intelligence methods.

Software Engineering for Artificial Intelligence: Data and Feature Engineering

Data and Feature Engineering: Toward Data Quality and Feature Engineering

From a practical perspective, there is not very much to say on the tools you can use to mine data: there exist a variety of instruments, but their use depends on what you would like to mine.

Much more can instead be said with respect to the follow up steps to conduct in order to make data qualitative enough to build reliable artificial intelligence methods.

Data Cleaning



Feature Construction



Feature Scaling



Feature Selection



Data Balancing

If the set of values of a variable is very much different from the set of values of another variable, an artificial intelligence algorithm might over/under-estimate the importance of one variable just because of the different distribution.

Feature Scaling. Set of techniques that enable the normalization/scaling of the values of a feature.

There are various methods already implemented in the most common data science toolkits. The most widely used are:

Min-Max Normalization

$$x' = a + \frac{(x - \min(x))(b - a)}{\max(x) - \min(x)}$$

Z-score Normalization

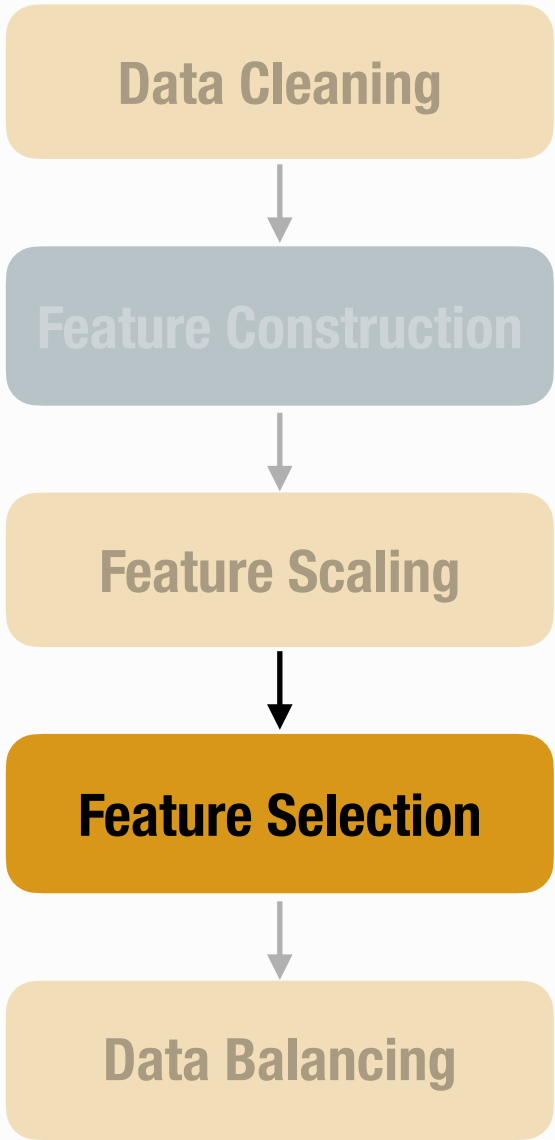
$$x' = \frac{(x - \bar{x})}{\sigma}$$

Software Engineering for Artificial Intelligence: Data and Feature Engineering

Data and Feature Engineering: Toward Data Quality and Feature Engineering

From a practical perspective, there is not very much to say on the tools you can use to mine data: there exist a variety of instruments, but their use depends on what you would like to mine.

Much more can instead be said with respect to the follow up steps to conduct in order to make data qualitative enough to build reliable artificial intelligence methods.



Feature Selection. Process through which the most relevant features are identified, while the non-relevant or redundant are filtered out.

While the feature construction process might identify features that might address the problem, it might also exist redundant features or features that decreases in presence of others.

While this feature might be relevant, what is its added value when the variable 'subject' is considered? It is likely a redundant feature!

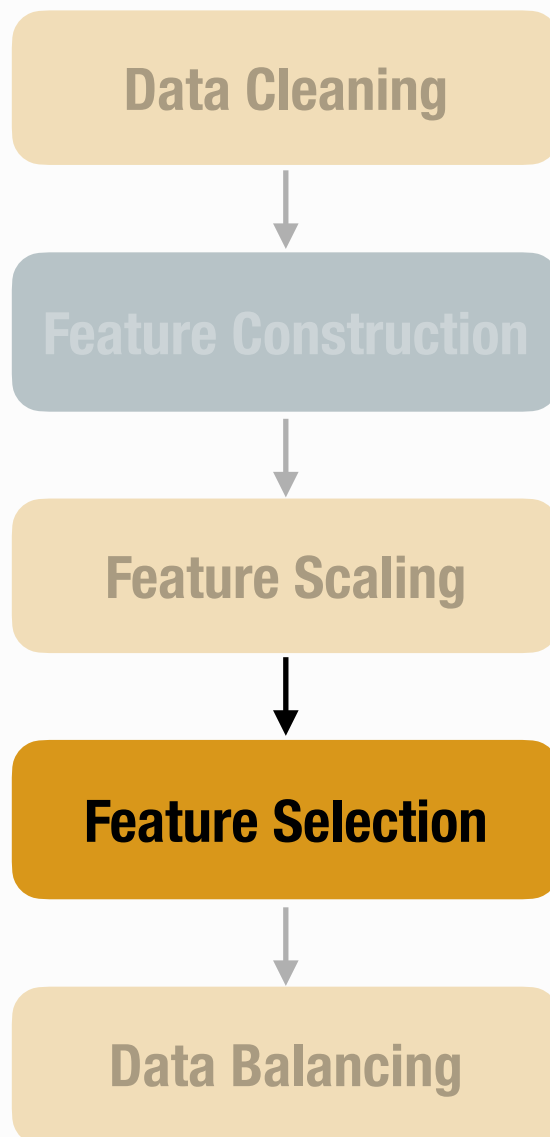
is_spam	domain	subject	attachments	length	text_color	contains_SE4AI
NO	<u>unisa.it</u>	[SE4AI] Exam	1	300	black	yes
NO	<u>gmail.com</u>	[SE4AI] Exam	-	180	black	yes
YES	hotmail.it	Business	-	120	black	no

Software Engineering for Artificial Intelligence: Data and Feature Engineering

Data and Feature Engineering: Toward Data Quality and Feature Engineering

From a practical perspective, there is not very much to say on the tools you can use to mine data: there exist a variety of instruments, but their use depends on what you would like to mine.

Much more can instead be said with respect to the follow up steps to conduct in order to make data qualitative enough to build reliable artificial intelligence methods.



Feature Selection. Process through which the most relevant features are identified, while the non-relevant or redundant are filtered out.

While the feature construction process has identified the features that might address the problem of interest, there might exist redundant features or even features whose relevance decreases in presence of other features.

Redundant features are problematic because they risk to bias the inner-working of artificial intelligence systems. If two variables provide the same or similar information, how can an automated technique decide which one to use for its goals?

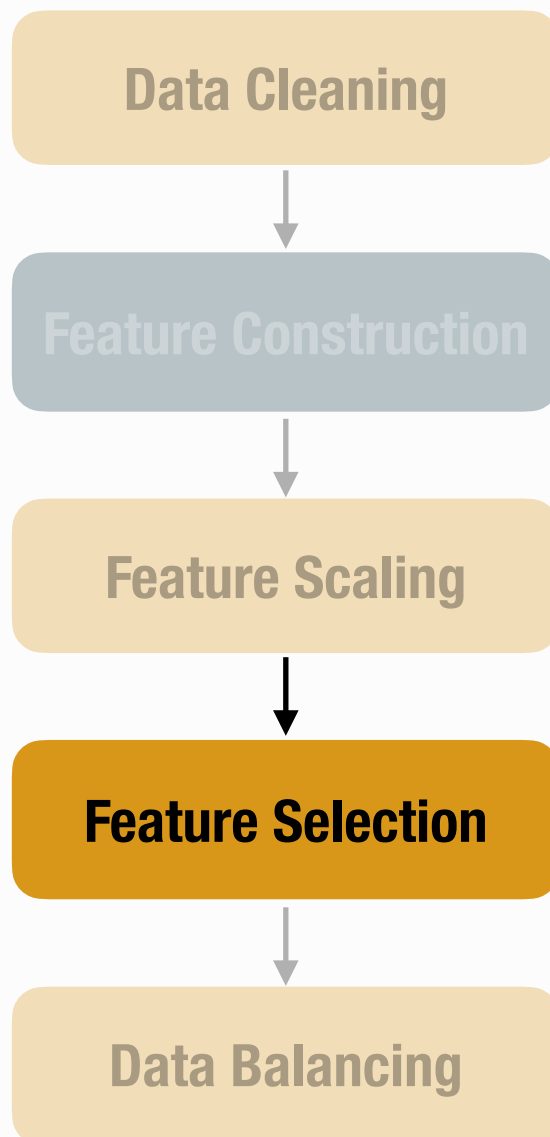
So, redundant variables must be identified and removed.

Software Engineering for Artificial Intelligence: Data and Feature Engineering

Data and Feature Engineering: Toward Data Quality and Feature Engineering

From a practical perspective, there is not very much to say on the tools you can use to mine data: there exist a variety of instruments, but their use depends on what you would like to mine.

Much more can instead be said with respect to the follow up steps to conduct in order to make data qualitative enough to build reliable artificial intelligence methods.



Feature Selection. Process through which the most relevant features are identified, while the non-relevant or redundant are filtered out.

Low-variance removal. The easiest way to proceed. You just remove the variables having a low data variance —> if all the values are similar, it is likely the variable does not discriminate the problem of interest.

Univariate feature removal. The method relies on statistical methods (e.g., the Pearson's correlation coefficient) to understand the correlation between pairs of variables. If two of them are highly correlated, one of them should be removed.

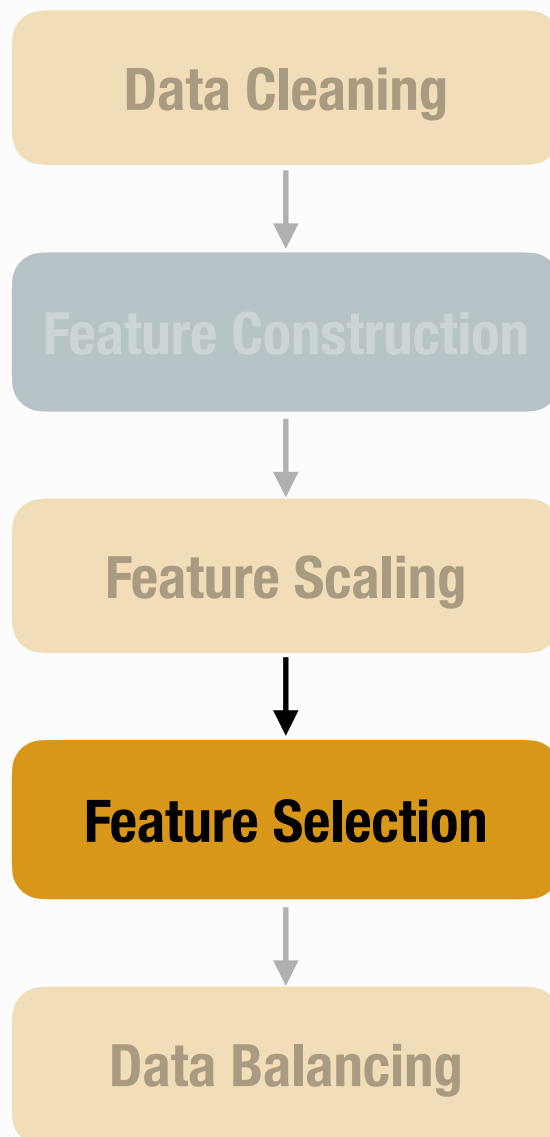
Please note that there exist other, plenty of other methods!

Software Engineering for Artificial Intelligence: Data and Feature Engineering

Data and Feature Engineering: Toward Data Quality and Feature Engineering

From a practical perspective, there is not very much to say on the tools you can use to mine data: there exist a variety of instruments, but their use depends on what you would like to mine.

Much more can instead be said with respect to the follow up steps to conduct in order to make data qualitative enough to build reliable artificial intelligence methods.



Feature Selection. Process through which the most relevant features are identified, while the non-relevant or redundant are filtered out.

In general, we can distinguish:

- (1) Unsupervised methods, like those seen before, that do not rely on information of the problem of interest to identify redundant variables.
- (2) Supervised methods, which create models that exploit the problem of interest to identify irrelevant variables that do not affect the performance of the artificial intelligence algorithm.

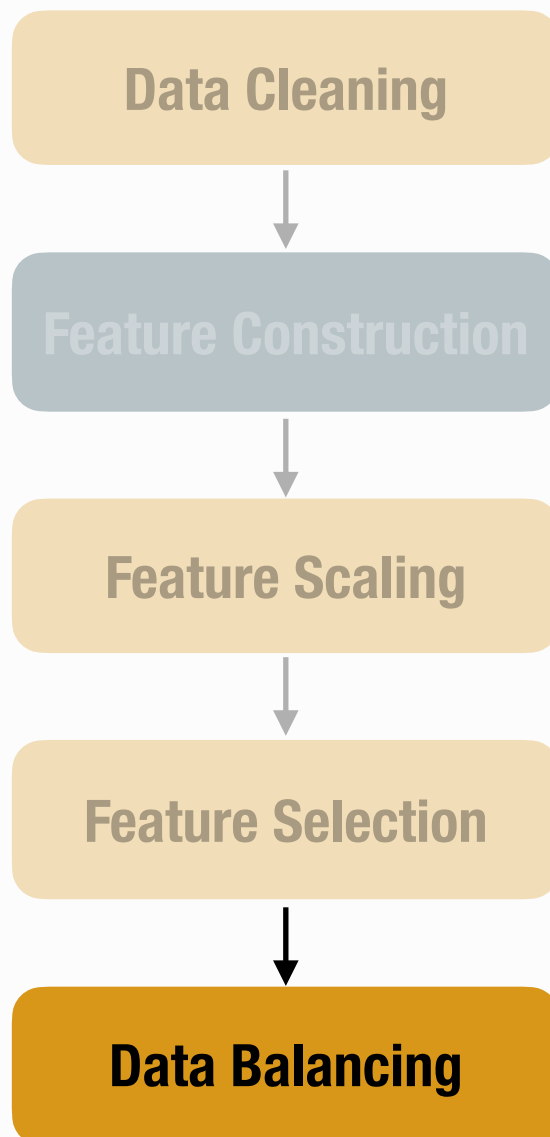
In the next lecture, we will talk of supervised methods; some of them may be used as a way to select features.

Software Engineering for Artificial Intelligence: Data and Feature Engineering

Data and Feature Engineering: Toward Data Quality and Feature Engineering

From a practical perspective, there is not very much to say on the tools you can use to mine data: there exist a variety of instruments, but their use depends on what you would like to mine.

Much more can instead be said with respect to the follow up steps to conduct in order to make data qualitative enough to build reliable artificial intelligence methods.



As we will further explore in the scope of the course, most artificial intelligence solutions work well only if they have enough data for all the classes involved in the problem.

For instance, how can an automated solution discriminate spam and no-spam emails if it does not have enough examples of no-spam emails? How can it distinguish the characteristics of the emails of the two classes?

In reality, most problems are *unbalanced*. From medicine to agriculture, we hardly have balanced datasets to use.

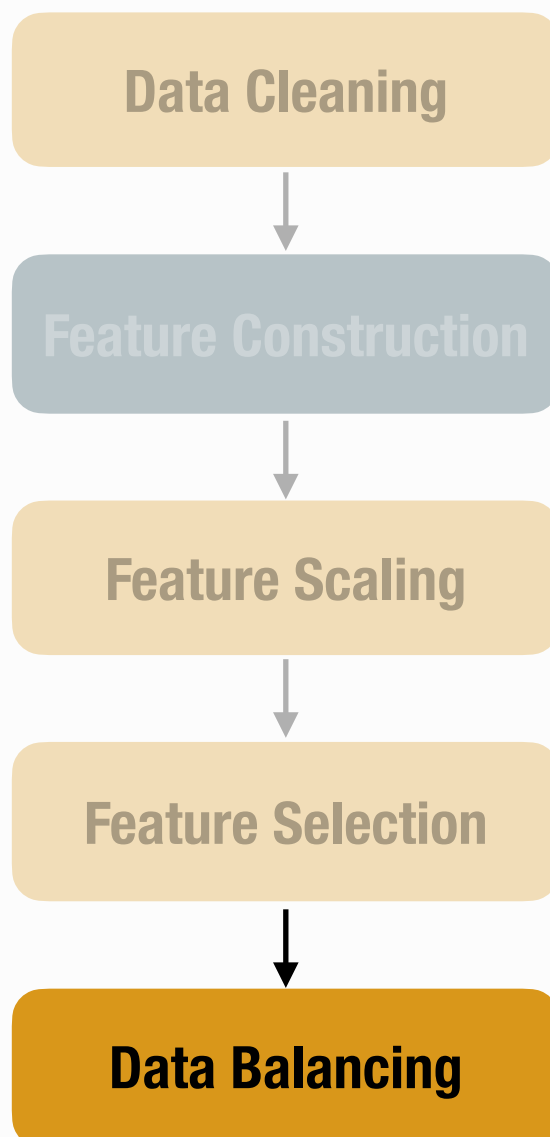
Data Balancing. Set of techniques that might convert an unbalanced dataset into a balanced one by means of analysis of the data available.

Software Engineering for Artificial Intelligence: Data and Feature Engineering

Data and Feature Engineering: Toward Data Quality and Feature Engineering

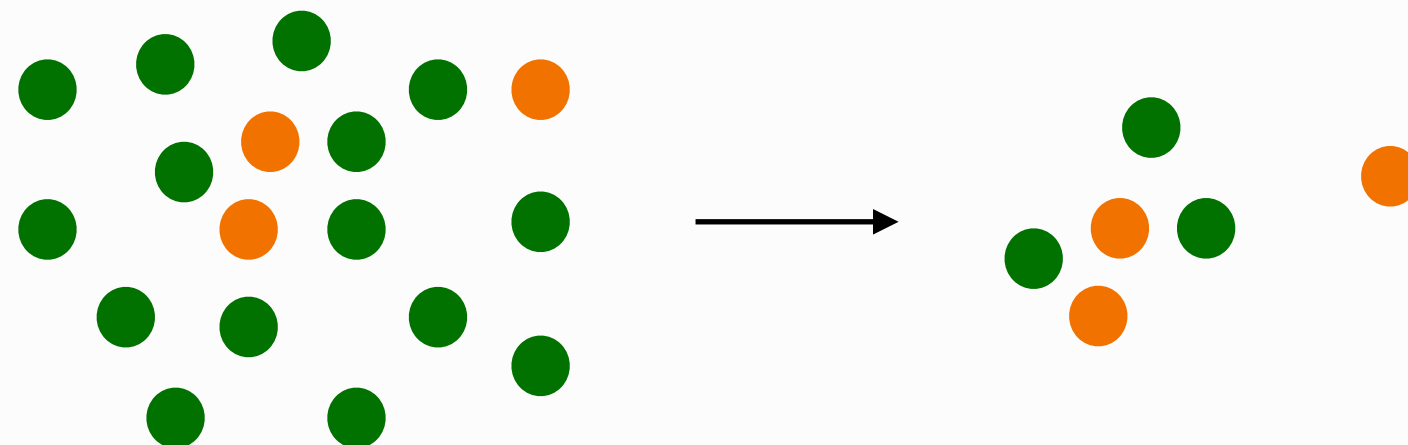
From a practical perspective, there is not very much to say on the tools you can use to mine data: there exist a variety of instruments, but their use depends on what you would like to mine.

Much more can instead be said with respect to the follow up steps to conduct in order to make data qualitative enough to build reliable artificial intelligence methods.



Data Balancing. Set of techniques that might convert an unbalanced dataset into a balanced one by means of analysis of the data available.

Undersampling. Pseudo-random removal of data from the majority class.



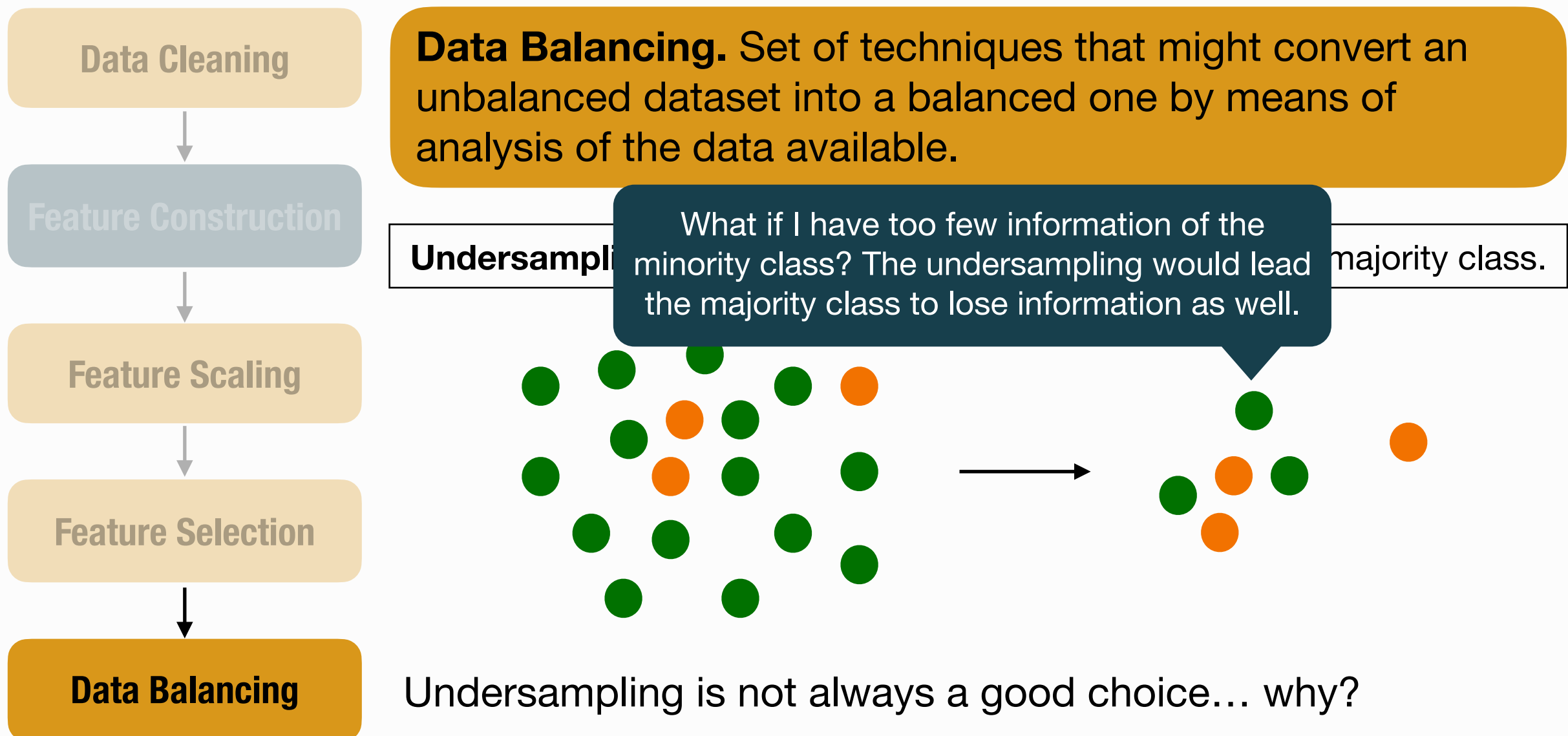
Undersampling is not always a good choice... why?

Software Engineering for Artificial Intelligence: Data and Feature Engineering

Data and Feature Engineering: Toward Data Quality and Feature Engineering

From a practical perspective, there is not very much to say on the tools you can use to mine data: there exist a variety of instruments, but their use depends on what you would like to mine.

Much more can instead be said with respect to the follow up steps to conduct in order to make data qualitative enough to build reliable artificial intelligence methods.

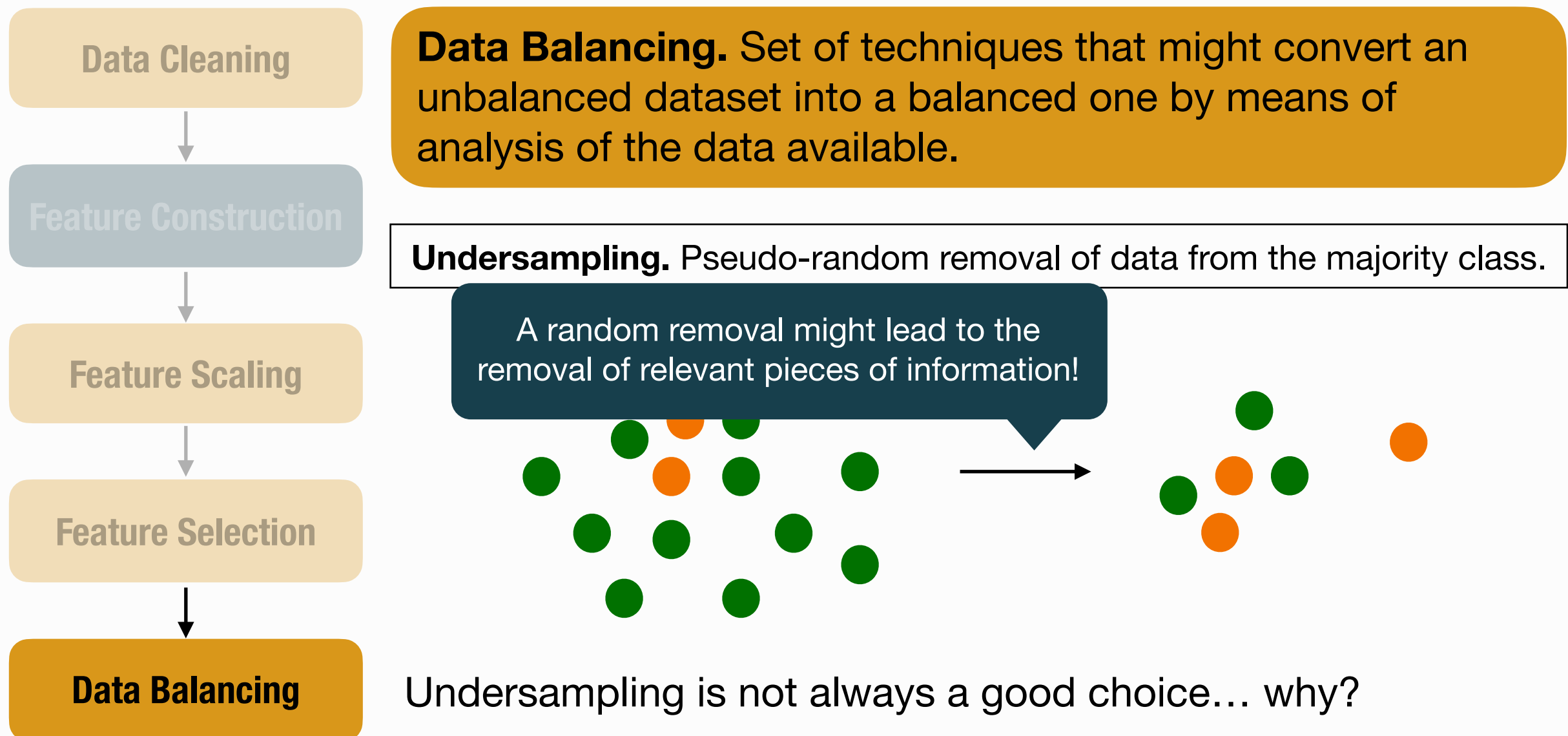


Software Engineering for Artificial Intelligence: Data and Feature Engineering

Data and Feature Engineering: Toward Data Quality and Feature Engineering

From a practical perspective, there is not very much to say on the tools you can use to mine data: there exist a variety of instruments, but their use depends on what you would like to mine.

Much more can instead be said with respect to the follow up steps to conduct in order to make data qualitative enough to build reliable artificial intelligence methods.

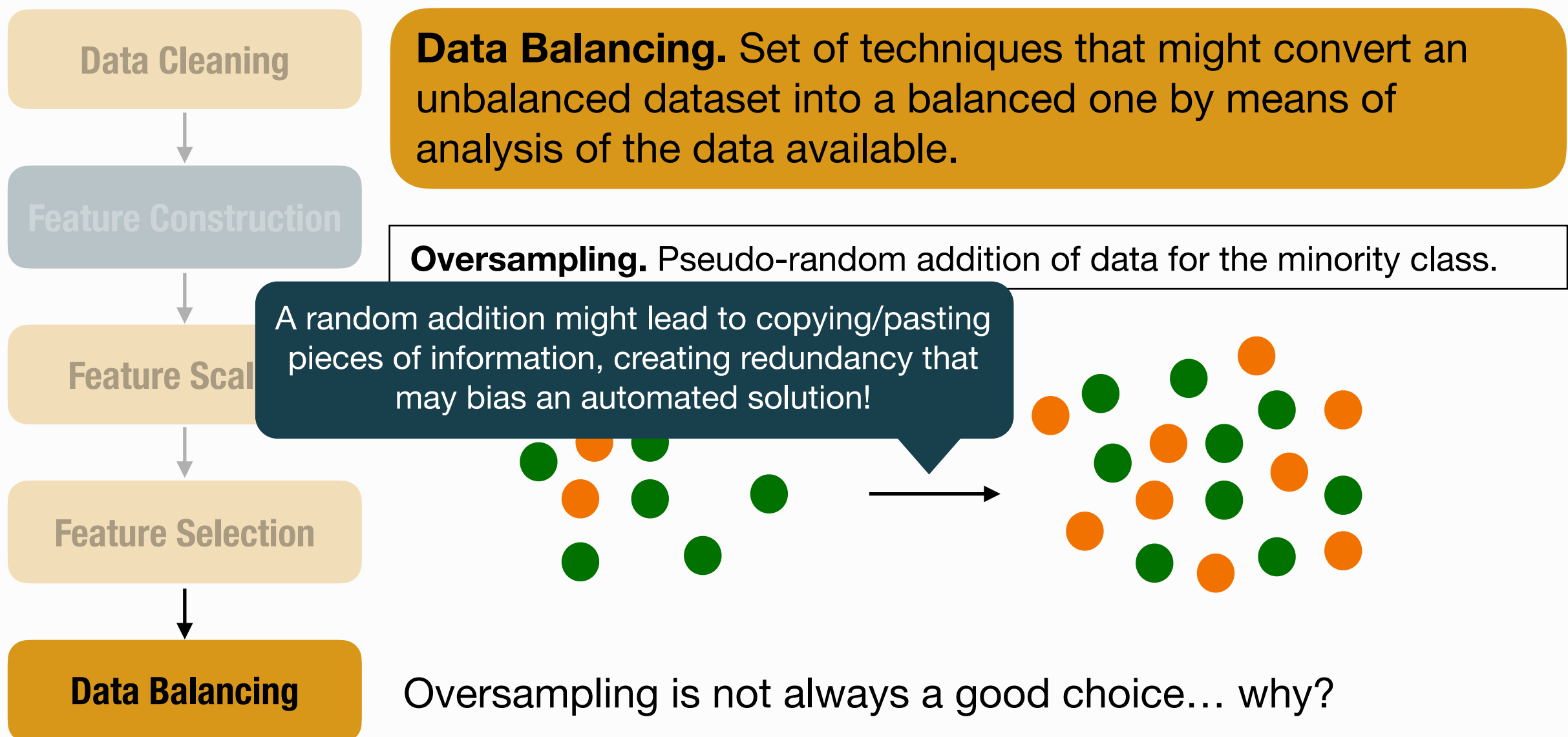


Software Engineering for Artificial Intelligence: Data and Feature Engineering

Data and Feature Engineering: Toward Data Quality and Feature Engineering

From a practical perspective, there is not very much to say on the tools you can use to mine data: there exist a variety of instruments, but their use depends on what you would like to mine.

Much more can instead be said with respect to the follow up steps to conduct in order to make data qualitative enough to build reliable artificial intelligence methods.

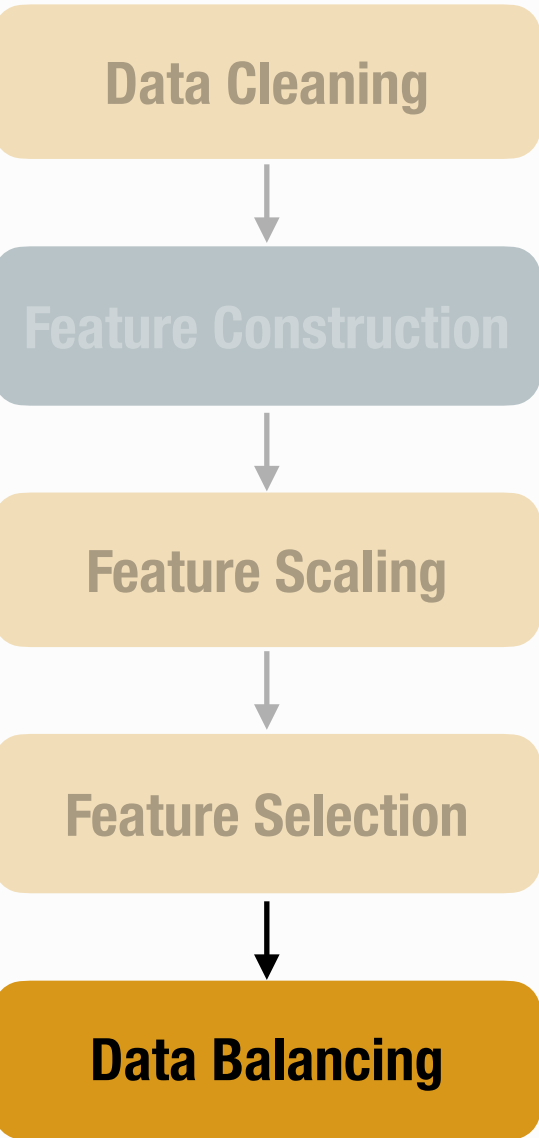


Software Engineering for Artificial Intelligence: Data and Feature Engineering

Data and Feature Engineering: Toward Data Quality and Feature Engineering

From a practical perspective, there is not very much to say on the tools you can use to mine data: there exist a variety of instruments, but their use depends on what you would like to mine.

Much more can instead be said with respect to the follow up steps to conduct in order to make data qualitative enough to build reliable artificial intelligence methods.



Data Balancing. Set of techniques that might convert an unbalanced dataset into a balanced one by means of analysis of the data available.

Can we do it better?

Generation of synthetic instances of the minority class through a four-step approach that combines clustering techniques and statistical methods.

SMOTE is the standard de-facto for balancing artificial intelligence data; since 2002, the paper has received more than 17,000 citations!

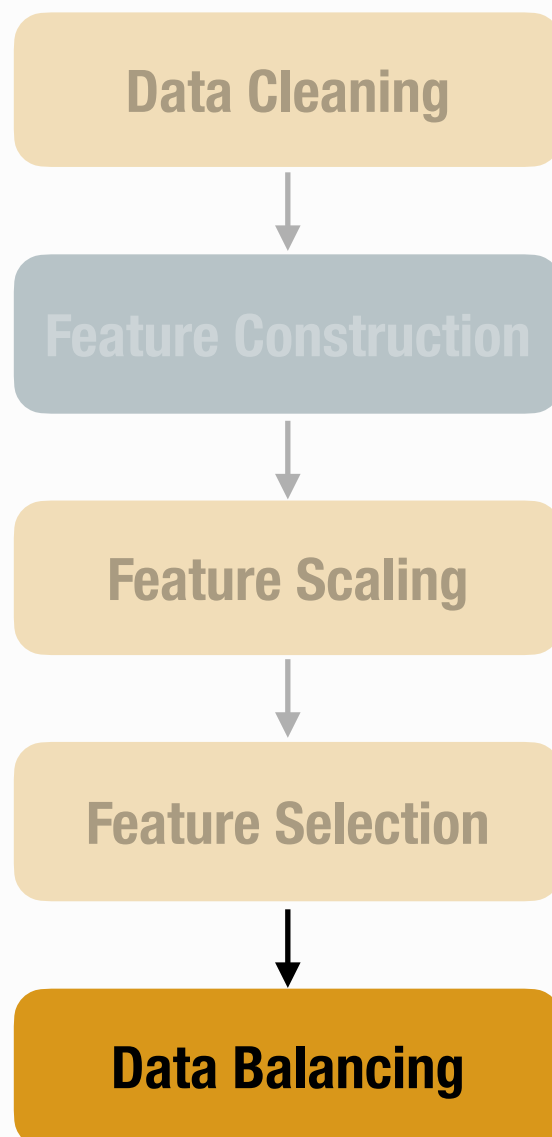


Software Engineering for Artificial Intelligence: Data and Feature Engineering

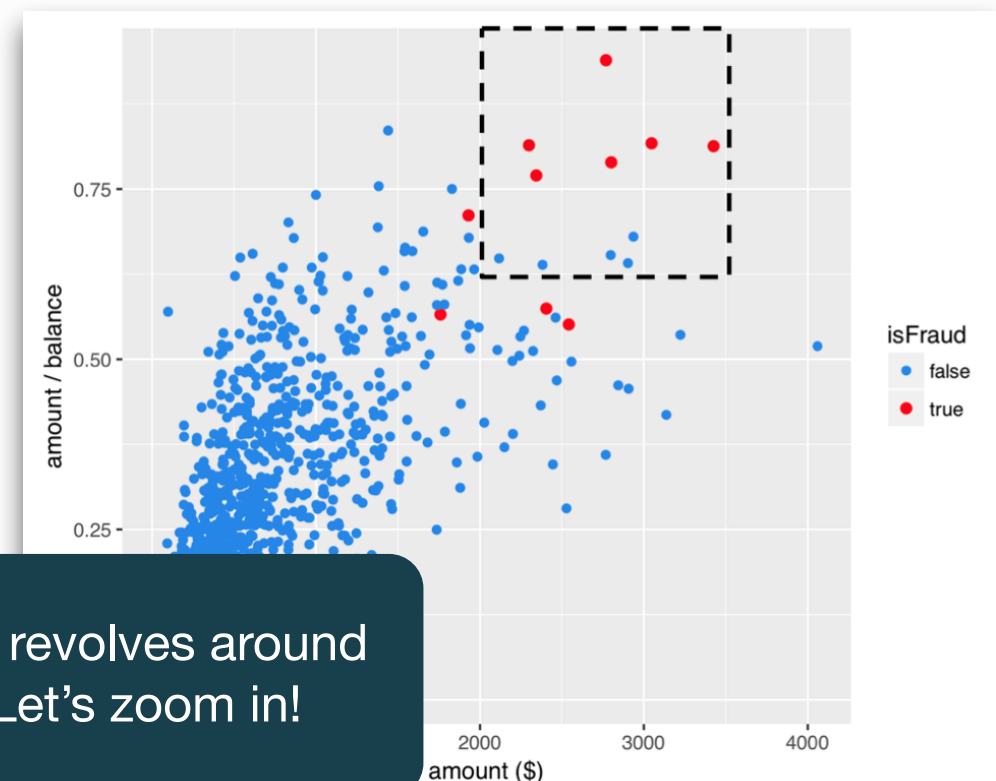
Data and Feature Engineering: Toward Data Quality and Feature Engineering

From a practical perspective, there is not very much to say on the tools you can use to mine data: there exist a variety of instruments, but their use depends on what you would like to mine.

Much more can instead be said with respect to the follow up steps to conduct in order to make data qualitative enough to build reliable artificial intelligence methods.



Data Balancing. Set of techniques that might convert an unbalanced dataset into a balanced one by means of analysis of the data available.



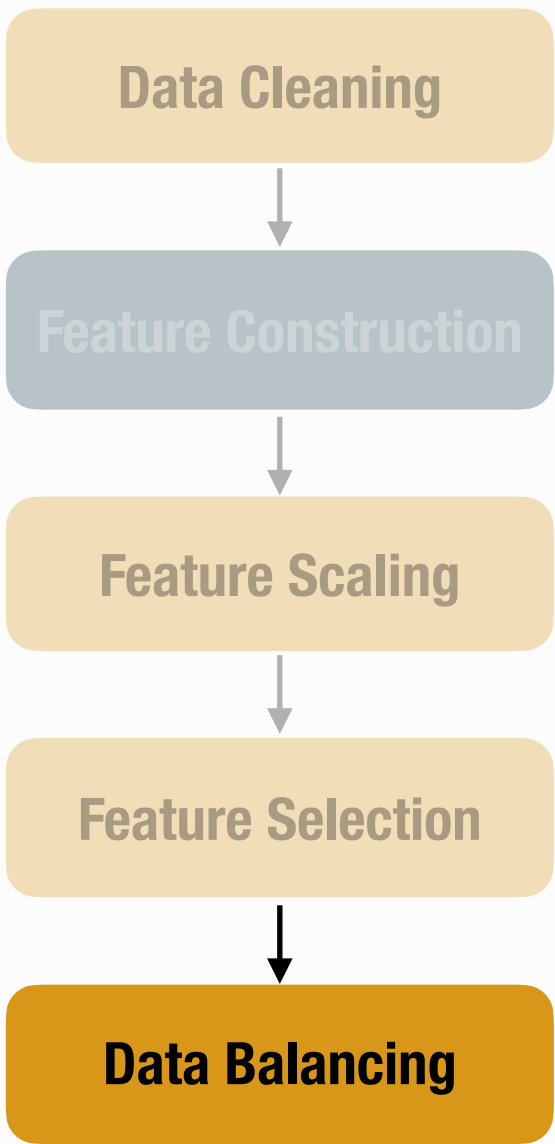
Let's consider this case, which revolves around the classification of frauds. Let's zoom in!

Software Engineering for Artificial Intelligence: Data and Feature Engineering

Data and Feature Engineering: Toward Data Quality and Feature Engineering

From a practical perspective, there is not very much to say on the tools you can use to mine data: there exist a variety of instruments, but their use depends on what you would like to mine.

Much more can instead be said with respect to the follow up steps to conduct in order to make data qualitative enough to build reliable artificial intelligence methods.



Data Balancing. Set of techniques that might convert an unbalanced dataset into a balanced one by means of analysis of the data available.

- Select a random instance X of the minority class.
- Find the k -nearest neighbors of X , e.g., 4.
- Randomly choice of the the k -nearest neighbors of X .
- Generate a synthetic instance.
- Repeat for all neighbors.

X_amount	X_ratio
2800	0.79

X4_amount	X4_ratio
2770	0.94

Choose random number in $[0;1]$, e.g., 0.6.
Scaling of the data based on this number.

$2.800 + 0.6 * (2770 - 2800)$	$0.79 + 0.6 * (0.94 - 0.79)$
Xnew_amount	Xnew_ratio
2782	0.88

Software Engineering for Artificial Intelligence: Requirements Engineering

Before leaving... What Should You Remember?

Software Engineering for Artificial Intelligence: Data and Feature Engineering

Data and Feature Engineering: Why?

There cannot be artificial intelligence without data. There cannot be accurate artificial intelligence without *good* data.

Data Governance. The management process that curates the availability, usability, integrity, and safety of the data.

Historically speaking, data governance has received increasing attention because of multiple reasons connected to financial, environmental, and governmental regulations. Nowadays, data represent one of the major values of companies, organizations, and governments. Implementing strategies to collect, curate, and store data is therefore a urgent need that pertain to the definition of internal and global standards.

The main focus of data governance is concerned with data quality —> good data have higher value and can be used to advance the state of the practice by means of AI!

Data Quality. The set of processes that describe the accuracy, completeness, and consistency of the data collected.

To enable data quality, we refer to data engineering.

Data Engineering. The set of techniques and algorithms that enable the extraction, analysis, and preparation of data. It is the backbone of artificial intelligence.

Software Engineering for Artificial Intelligence: Data and Feature Engineering

Data and Feature Engineering: Data Versus Features - Exercise

Suppose that you would like to design an automated solution able to forecast the number of students who will attend the next SE4AI lecture.

Which data would you use?

Raw data coming from the MS Teams reports

MS Teams collects statistics on the student's attendance for each lecture scheduled. This sounds like the ideal dataset to use! But is this the only available data?

Raw data coming from the course/exam schedule

Attendance might depend on other factors that you can measure, like the co-occurrence of institutional events. Raw data can be extracted by mining EasyLesson!

Which feature(s) would you build?

Average number of students per lecture?

Number of students attending the last lecture?

Presence of other institutional events?

Additional factors coming from your knowledge?

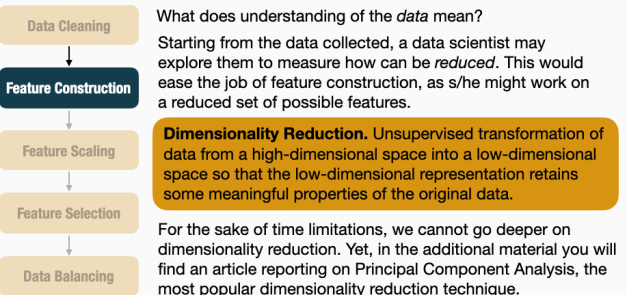
In other words, feature engineering is a creative process that involves the combination of data elaboration and domain knowledge!

Software Engineering for Artificial Intelligence: Data and Feature Engineering

Data and Feature Engineering: Toward Data Quality and Feature Engineering

From a practical perspective, there is not very much to say on the tools you can use to mine data: there exist a variety of instruments, but their use depends on what you would like to mine.

Much more can instead be said with respect to the follow up steps to conduct in order to make data qualitative enough to build reliable artificial intelligence methods.



Software Engineering for Artificial Intelligence: Data and Feature Engineering

Data and Feature Engineering: Why?

Data debt, uhm? A term which is borrowed from the agile software development vocabulary and revolves around the application of sub-optimal data quality and management processes that lead data not to be meaningful enough.

Data debt is the major obstacle to data-driven organizations. This includes various sources of problems with data.

- Structural debt** Quality issue with the design of a table, column, or view. Examples are the improperly named column, insufficiently normalized data, etc.
- Data quality debt** Quality issue with the consistency or usage of data values. Examples are the duplicated business key value, corrupted data, etc.
- Integrity debt** Quality issue with whether a referenced row exists within another table and that a row which is no longer needed is (soft) deleted appropriately, e.g., null foreign keys.
- Architectural debt** Quality issue with how external programs interact with a data source. Examples are inappropriate security access control, inappropriate encapsulation, etc.
- Documentation debt** Quality issue with with any supporting documents, including models. Examples are inconsistent or outdated information.
- Functional debt** Quality issue with execution aspects within a data source, such as stored procedures, stored functions, and triggers.

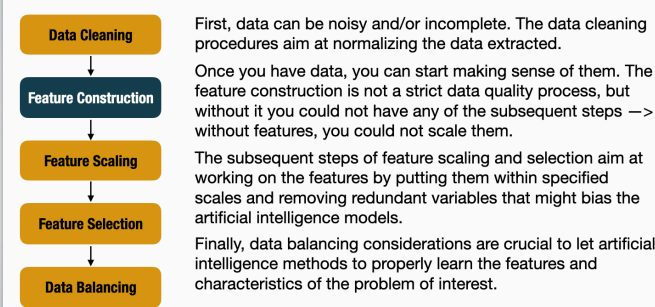
While some of these issues can be tackled with good database skills, some are concerned with the preprocessing steps during the data extraction phase.

Software Engineering for Artificial Intelligence: Data and Feature Engineering

Data and Feature Engineering: Toward Data Quality and Feature Engineering

From a practical perspective, there is not very much to say on the tools you can use to mine data: there exist a variety of instruments, but their use depends on what you would like to mine.

Much more can instead be said with respect to the follow up steps to conduct in order to make data qualitative enough to build reliable artificial intelligence methods.

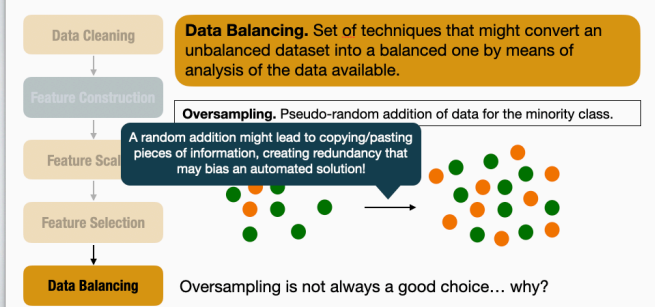


Software Engineering for Artificial Intelligence: Data and Feature Engineering

Data and Feature Engineering: Toward Data Quality and Feature Engineering

From a practical perspective, there is not very much to say on the tools you can use to mine data: there exist a variety of instruments, but their use depends on what you would like to mine.

Much more can instead be said with respect to the follow up steps to conduct in order to make data qualitative enough to build reliable artificial intelligence methods.



Software Engineering for Artificial Intelligence

Data and Feature Engineering

